

Industrial AI Copilot

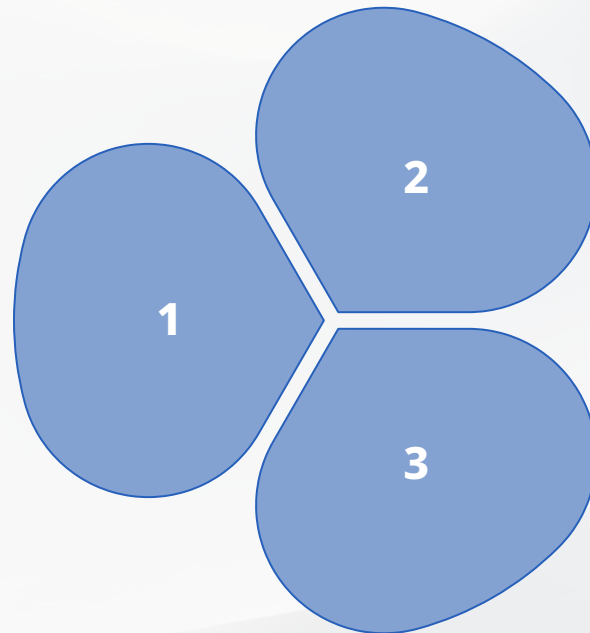
Predictive Maintenance & Anomaly Detection System for Critical Infrastructure — A Portfolio-Differentiating AI Engineering Project

PROBLEM STATEMENT

Why Critical Infrastructure Needs AI

Massive Data Scale

AWS data centers, nuclear facilities, fabs, and energy systems generate millions of sensor measurements continuously



Failures Are Costly

Server downtime, equipment damage, production losses, and safety risks make every missed failure extremely expensive

Reactive vs. Predictive

Traditional maintenance waits for failure. AI detects patterns before failure occurs, enabling preventive action

Four-Layer System Architecture

End-to-End Pipeline: Data → ML → AI → Deployment

A modular, production-grade architecture that demonstrates full-stack AI engineering capability.



Data Engineering Layer

Data cleaning, feature engineering, ETL pipeline from raw sensor streams



Machine Learning Layer

Random Forest, XGBoost, Isolation Forest, LSTM
Autoencoder



AI Engineering Layer

FastAPI backend, AI Decision Engine, RAG Knowledge Assistant



Deployment Layer

Docker, AWS, Monitoring Dashboard, Power BI / Streamlit

DATA SCIENCE

EDA & Feature Engineering



Sensor EDA

Analyze temperature, pressure, vibration, power, humidity, cooling, and equipment state signals



Engineered Features

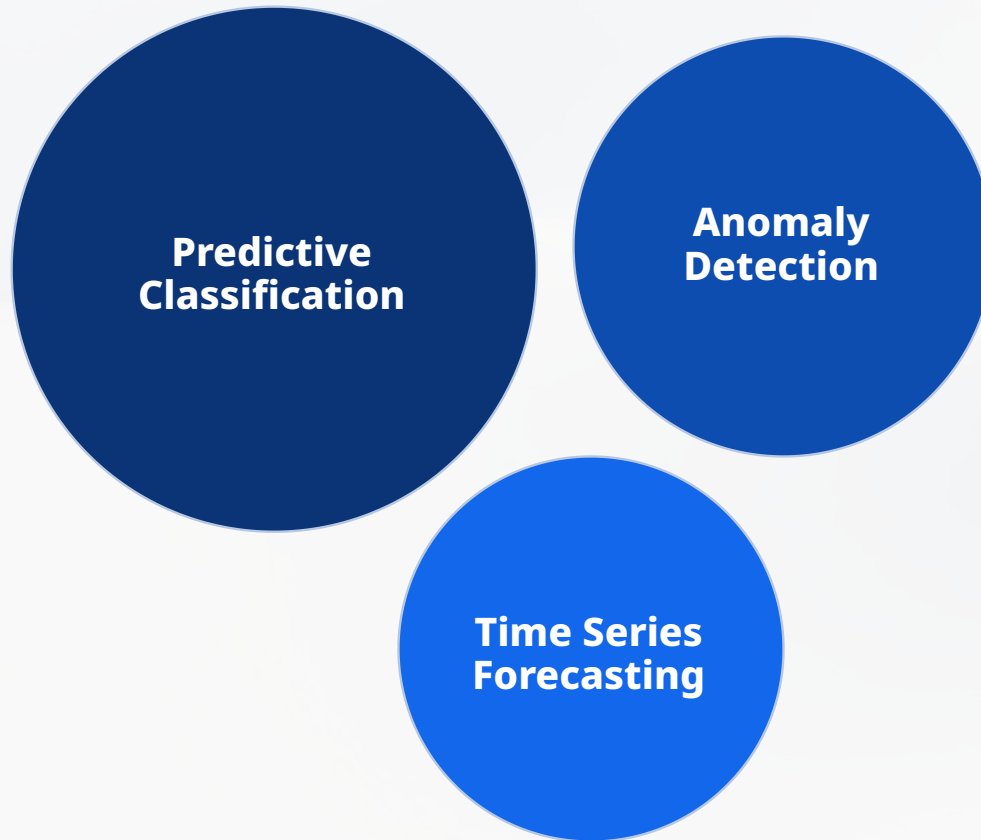
Temperature deviation, Equipment Health Score, and Failure Probability derived from raw sensor readings



Operational Intelligence

Combine signals: Temp ↑ + Fan speed abnormal + Power ↑ = Potential cooling failure detected

Three Machine Learning Models



Key Engineering Decisions

Recall is the critical metric — missing a failure is far more costly than a false alarm

Isolation Forest catches unknown failure modes no labeled data could predict

LSTM forecasts next 24-hour temperature trend and failure probability

Model ensembling combines classification and anomaly signals for robust alerts

Move beyond accuracy to optimize for operational safety. Precision, Recall, and F1 are all tracked in MLflow.

AI Operations Assistant — The Differentiator

01

LLM + RAG Core

LangChain + vector database (FAISS/Chroma) grounds the assistant in live sensor data and maintenance history

03

Actionable Recommendations

System outputs ranked corrective steps: inspect filter, verify airflow, review maintenance history

02

Natural Language Queries

"Why is Cooling Unit 3 showing abnormal behavior?" — the AI responds with sensor evidence and ranked causes

04

FastAPI Backend

Production-ready REST API exposes predictions, anomaly scores, and AI assistant responses to any client



TECH STACK & REPO

Technology Stack & Repository Structure

Production-Grade Tools Across Every Layer

A recruiter-ready GitHub repo at [industrial-ai-copilot/](#) with clean separation of concerns across data, models, API, chatbot, and dashboard.

Data Science

Python · Pandas · NumPy ·
Scikit-learn · PyTorch

AI Engineering

LangChain · Hugging Face
· OpenAI API · FAISS /
Chroma

Backend & Deploy

FastAPI · Docker · AWS ·
Streamlit · Power BI

MLOps

MLflow experiment tracking
· GitHub Actions CI/CD
pipeline

Portfolio Impact & Career Outcomes

What This Project Signals to Recruiters



Full AI Engineering Stack

ML models + data pipelines + cloud deployment + APIs + LLM integration — all in one project.



High-Value Role Targeting

Supports applications for AI Engineer (\$150k–\$250k+), ML Engineer (\$160k–\$280k+), and AI Solutions Architect (\$180k–\$300k+).



Real-World Domain Impact

Energy grids, semiconductor fabs, data centers, and manufacturing — the highest-stakes AI frontier.



Beyond "I Trained a CNN"

Demonstrates engineering maturity, business impact thinking, and autonomous AI system design.

Data → AI reasoning → Automated decision support. This is the new paradigm — and this project proves you can build it.